higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(Second Paper)
NQF LEVEL 3

22 FEBRUARY 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from a table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitute into formula
R/J	Reasoning/Justification
P	Penalty – no units, incorrect rounding off
R	Rounding off
E	Explanation
U	Unit
MF	Manipulation of Formula

This marking guideline consists of 9 pages

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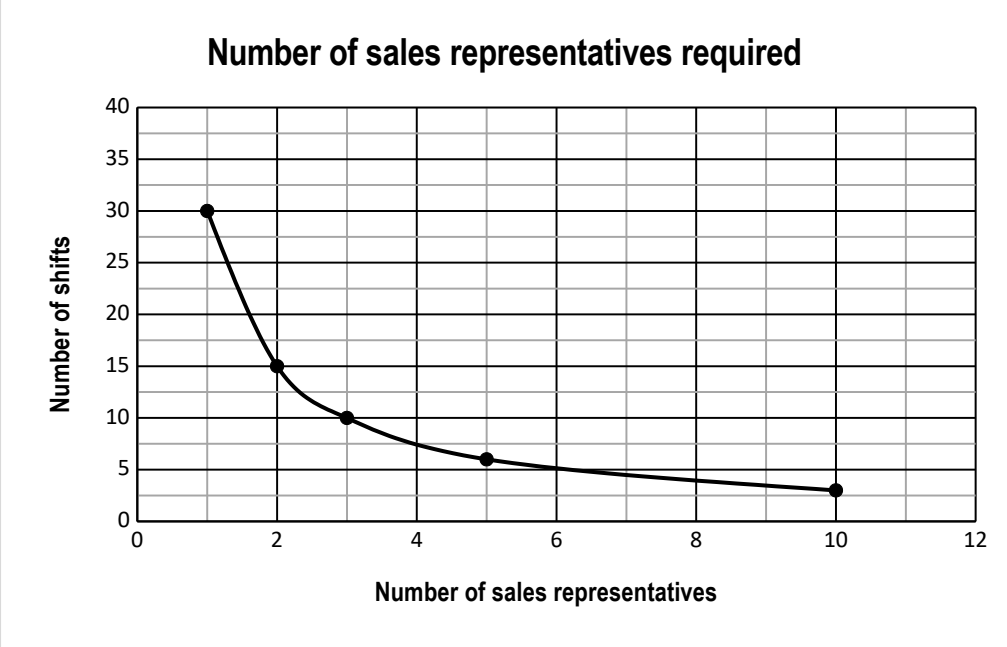
QUESTION 1 [40] *Do NOT penalise if units are omitted unless marks are allocated.		
Question	**Solution**	**Explanation**
1.1	Area of right angled triangle = 0,5 × 20✓ × 60✓ = 600✓ cm^2 Area of isosceles triangle = 0,5 × 20✓ × 40✓ = 400 ✓cm^2 Total area of off-cuts = 2(600)✓ + 400✓ = 1 600 cm^2✓	2SF 1 A 2SF 1A 2M 1A with unit (9)
1.2	Area of block = 60✓× 60✓ Area of block = 3 600 cm^2✓ Therefore, area of logo = 3 600 cm^2✓ – 1 600 cm^2✓ = 2 000 cm^2✓	2M 1A 2M 1CA (Q1.1) (6)
1.3 1.3.1	AB = $\sqrt{60^2 + 20^2}$ ✓ = 63,25✓ cm **(answer only full marks)**	1M/SF 1A (2)
1.3.2	CD = $\sqrt{40^2 + 10^2}$ ✓ = 41,23✓ cm **(answer only full marks)**	1M/SF 1A (2)
1.4	Total surface area of logo: = 2✓ (2 000)✓ + 3✓(20 × 10)✓ + 2✓ (63,25 × 10)✓ + 2✓(41,23 × 10)✓ = 6 689,6 cm^2✓	 2MA 2(2 000) 2MA + 3(20 × 10) 2M +2(63,25 × 10) 2M +2(41,23 × 10) 1 CA (9)
1.5	Using area: Percentage of wood = $\frac{2\,000}{3\,600}$ ✓×100%✓ = 55,56% Or Using Volume: Percentage of wood = $\frac{20\,000}{36\,000}$ ✓×100%✓ = 55,56✓%	 2M 1A 2M 1A (3)

1.6	Number of tins = 30✓× 6 689,6✓÷ 5 000✓ = 40,1376✓ = 41✓	3M 1A 1 Round up (5)
1.7	Volume of Sphere = $\frac{4}{3}\times 3{,}14\checkmark\times 31^3$ ✓✓ = 124 724,99 cm^3✓	1A radius =31 2SF 1A with unit (4)
		[40]

QUESTION 2 [40] *Do NOT penalise if 'R' is omitted.		
Question	**Solution**	**Explanation**
2.1.1	To pay income tax in advance✓ Or To pay preliminary income tax✓	1E (1)
2.1.2	UIF contribution(B) = R22 611,00✓ × 0,01✓ = R226,11✓ **(answer only full marks)**	2M 1A (3)
2.1.3	A = R22 611,00 + R12 360,98 = R34 971,98✓ C = 6 827,54+1 695,87 + 226,11 + 4 500,00 + 450,00 + 35,00 = R13 734,52✓ C = R34 971,98✓– R13 724,52✓ C = 21 237,46✓	1A 1A 2SF 1A (5)
2.1.4	Increase = $\frac{\text{R}24\,193{,}77\checkmark-\text{R}22\,611{,}00\checkmark}{\text{R}22\,611{,}00\checkmark}\times 100\checkmark$ = 7%✓	3M 1RT 1A (5)
2.2 2.2.1	Loan amount = R200 000 ✓× 0,9%✓ Loan amount = R180 000✓ **(answer only full marks)**	2M 1A (3)
2.2.2	Number of thousands = 180 000 ÷ 1 000✓ = 180✓ **(answer only full marks)**	1M 1A (2)
2.2.3	Monthly repayment = 24,89 ✓× 180✓+ R57,00✓ = R4 537,20✓ **(3 marks for R4 537,2)**	1RT 24,89 1M × 180 1M + R57,00 1A (4)

2.2.4	Option 1 = 25,89✓× 180 ✓ = R4 660,20✓ **(2 marks for R4 60,2) – Penalise once only** Total cost = R4 660,20 × 48✓ Total cost = R223 689,60✓ Option 2 = 20,07✓× 180 ✓ = R3 612,60✓ Total cost = R3 612,60 × 72✓ Total cost = R260 107,20✓ Difference: R260 107,20 – R223 689,60✓= R36 417,60✓	 1RT 25,89 1M × 180 1A 1M 1A 1RT 20,07 1M × 180 1A 1M 1A 1M 1A (12)
2.3	R460 = VAT exclusive price✓ × 115%✓ 1,15 ✓× VAT exclusive price = 460 VAT exclusive price = 460 ÷ 1,15✓ VAT exclusive price = R400✓ Or Let the original price be x R460 = x✓ × 115%✓ 1,15✓x = 460 x = 460 ÷ 1,15✓ x = R400✓	2M 1A 1,15 1MF 1A 2M 1A 1,15 1MF 1A (5)
		[40]

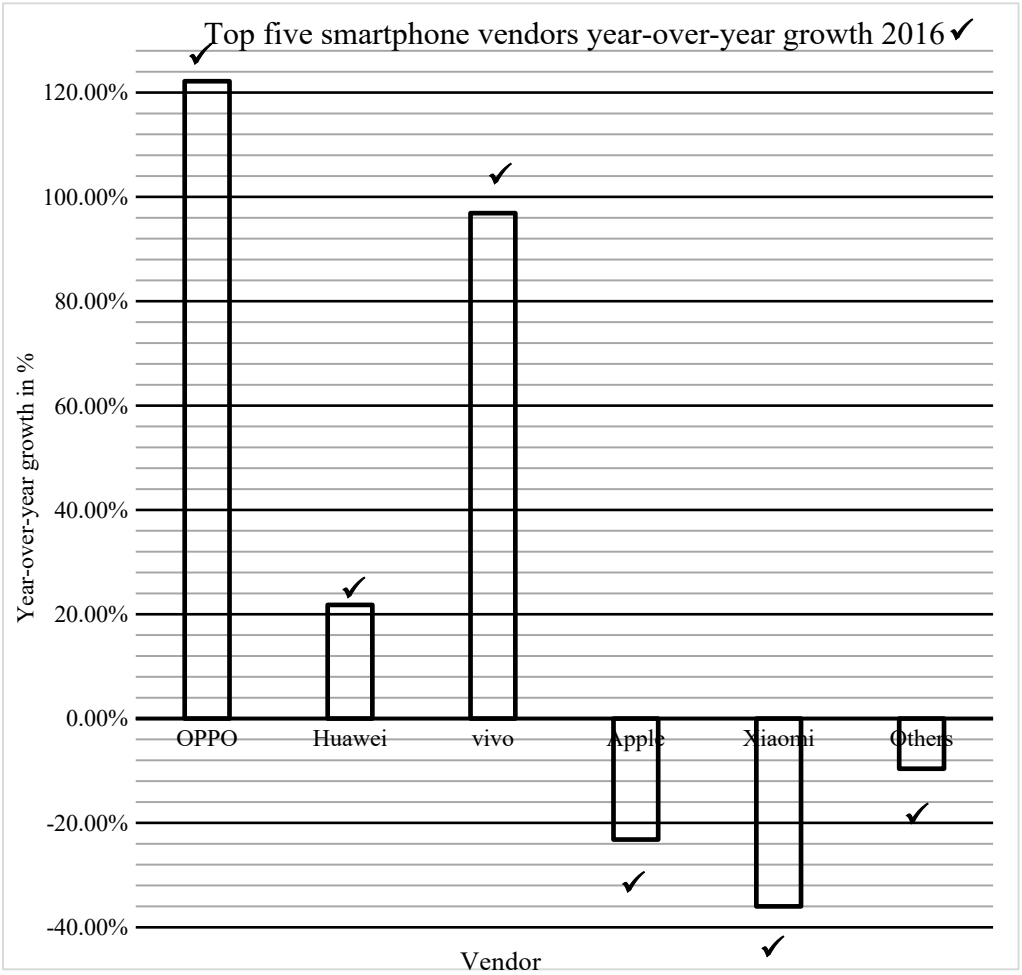
QUESTION 3 [35] *Do NOT penalise if 'R' is omitted.			
Question	**Solution**		**Explanation**
3.1 3.1.1	Fixed monthly running cost= R30 000✓✓		2RG (2)
3.1.2	Average cost = (60 000✓– 30 000✓) ÷ 20✓ = R1 500	Average cost = (90 000✓– 30 000✓) ÷ 40✓ = R1 500	2RG 1MA (3)
3.1.3	Average total monthly expenses = R30 000✓ + (R1 500✓ × no. of smartphones✓) **Or** = R30 000✓ + (R1 500✓ × n✓)		3A 3A (3)
3.1.4	average selling price of a smartphone = R40 000 ÷ 20✓ = R2 000✓	average selling price of a smartphone = R80 000÷ 40✓ = R2 000✓	1RG 1A (2)
3.1.5	average total monthly income = R2 000✓ × no. of smartphones✓ **Or** = R2 000✓ × n✓		2A 2A (2)
3.1.6	61✓✓ **(No marks for 60)**		2RG (2)
3.1.7	Profit for sale of 100 smartphones = R200 000 ✓– R180 000✓✓ =R20 000✓ **(answer only full marks)**		2RG and 1SF 1A (4)

3.1.8	Using the graphs: Profit = R220 000✓ – R195 000✓✓ Profit = R25 000✓ Candidate extended the lines✓ Using formulae: Cost = R30 000 + 110 ✓× R1 500 Cost = R195 000✓ Income = 110 × R2 000 Income = R220 000✓ Profit = R220 000 – R195 000✓ Profit = R25 000✓	 2RG and 1SF 1A 1M (award 1 mark although no evidence. The correct answer serve as evidence. No mark if answer is incorrect) 1SF 1A 1A 1SF 1A (5)
3.2 3.2.1	Constant value = 30✓✓	2RG (2)
3.2.2	A = 6✓✓ B = 3✓✓	2A 2A (4)
3.2.3	 Marks: 3 correct plotting of 1st 3 points✓✓✓ 2 for plotting the other 2 points✓✓ (2CA) 1 correct smooth line ✓	(6)
		[35]

QUESTION 4 [35]		
Question	**Solution**	**Explanation**
4.1 4.1.1	Smartphones sold = 15 +39 + 12 +8✓ = 74✓ (answer only full marks)	1M 1A (2)
4.1.2	Probability(Hisense) = $\frac{12✓}{74✓} = \frac{6}{37}✓$	1A numerator 1A denominator 1 CA fraction (3)
4.1.3	Probability(Samsung) = $\frac{15✓}{74✓} = 20,27\%✓$	1A numerator 1A denominator 1CA percentage (3)
4.2 4.2.1	35,3✓ million✓ Or 35,3✓× 1 000 000 = 35 300 000✓	1RT 1A 1RT 1A (2)
4.2.2	Difference = 76,6✓ – 62,9 ✓ million = 13,7 million or 13 700 000✓	2RT 1A (3)
4.2.3	Xiaomi✓✓	2RT (2)
4.2.4	Market volume Apple 2015 = 58,4 ÷ 429,9✓ × 100%✓ = 13,584 ≈ 13,6%	1RT 1M (2)
4.2.5	Xiaomi = $\frac{41,5✓ - 64,9✓}{64,9✓} \times 100✓$ year-on-year growth = - 36,055 ≈ - 36,1%	 2RT 2M (4)

4.2.6	**2016 Shipment Volume (units in millions)** 41.5✓ 44.9 69.2 76.6 Median = $\frac{69.2+76.6}{2}$ ✓= 72.9✓ 78.4 156.7 **2015 Shipment Volume (units in millions)** 35.1✓ 35.3 58.4 62.9 Median = $\frac{58.4+62.9}{2}$ = 60.65✓ 64.9 173.3 Difference = 72.9 – 60.65 = 12,25✓	1A ascending order 1M 1A 1A ascending order 1A 1CA (6)
4.2.7	• They may form part of 'other'✓ • They may have NOT been shipped from China✓ • Their shipment volumes may be very low and negligible✓ (Any other appropriate answer)	1RJ (1)

4.2.8



Marks:

6 correct bars✓✓✓✓✓✓
1 suitable heading✓

(7)

[35]

TOTAL: 150

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